

Contact Information

Mathematical Sciences Building, Office 609
Department of Mathematics, Purdue University

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Research Interests

Dynamical Systems (e.g. Mathematical Physics, Symplectic Geometry, Modeling in Material Science), Machine Learning (e.g. Physics-Informed Neural Networks), Computational & Applied Mathematics (e.g. Numerical Methods, Mathematical Algorithms, Data Assimilation).

Education & Background

Doctor of Philosophy in Mathematics

May 2029

Purdue University, West Lafayette, Indiana | Advisor: Prof. Aaron Yip

Master of Science in Mathematics

May 2026

Purdue University, West Lafayette, Indiana

Bachelor of Science in Pure Mathematics (*Magna Cum Laude*)

December 2022

Curricular Sequence in Applied Mathematics for Science and Engineering

University of Puerto Rico, Mayagüez Campus (UPRM), Mayagüez, Puerto Rico | Mentor: Reyes M. Ortiz Albino

Skills and Other Information

Programming & Computation: Python, Julia, C++, SageMath, MATLAB | **Formatting & Tools:** HTML, $\LaTeX$, Git
Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | **Spoken Languages:** English and Spanish

Research Experience

URA-Sandia Graduate Student Summer Fellow

URA-Sandia Graduate Student Summer Fellowship (administered by ORISE and ORAU)
Computational & Information Sciences Foundation, Sandia National Laboratories
Supervised by: Dr. Moe Khalil & Dr. Rileigh Bandy, Sandia National Laboratories
Research Topics: Machine Learning-based Surrogate Models, Data Assimilation

May 2025–August 2025
Livermore, California

GEM Fellow Summer Research Program Intern

MIT Lincoln Laboratory Summer Research Program (GEM Fellowship Employer Sponsor)
Group 39, Division 3, MIT Lincoln Laboratory, Massachusetts Institute of Technology
Supervised by: Dr. Sam Polk & Dr. Mabel Ramírez, MIT Lincoln Laboratory
Research Topics: Classical Machine Learning, Bayesian Inference, Generative AI, Unsupervised Learning

May 2023–August 2023
Lexington, Massachusetts

Research Opportunity for Undergraduate Students in STEM

Puerto Rico Louis Stokes Alliance for Minority Participation
Department of Mathematical Sciences, University of Puerto Rico, Mayagüez Campus
Supervised by: Prof. Reyes M. Ortiz Albino, University of Puerto Rico at Mayagüez
Research Topics: Number Theory, Commutative Algebra, Factorization Theory in Integral Domains

August 2019–December 2022
Mayagüez, Puerto Rico

NSF REU Participant

- Summer@ICERM 2022: Computational Combinatorics
Institute for Computational and Experimental Research in Mathematics, Brown University
Supervised by: Prof. Pamela E. Harris, University of Wisconsin-Milwaukee
Research Topics: Enumerative Combinatorics, Parking Functions
- NSF REU in Combinatorics, Probability and Algebraic Coding Theory
East Tennessee State University & University of Puerto Rico at Ponce
Supervised by: Prof. Fernando Piñero González, University of Puerto Rico at Ponce
Research Topics: Algebraic Geometry & Linear Error Correcting Codes

June 2022–August 2022
Providence, Rhode Island

June 2021–August 2021
Johnson City, Tennessee

Projects

Project in Numerical Partial Differential Equations

Purdue University, West Lafayette
MA61500: Numerical Methods for Partial Differential Equations | Instructor: Prof. Di Qi
TBA

January 2026–May 2026

Project in Numerical Differential Equations

Purdue University, West Lafayette

August 2025–December 2025

MA57300: Numerical Solutions of Ordinary Differential Equations | Instructor: Prof. Di Qi

Data Assimilation for the Lorenz 96 Model

Projects in Optimal Transport and Neural Networks

Purdue University, West Lafayette

January 2025–May 2025

MA59500OT: Computational Optimal Transport and Deep Generative Models | Instructor: Prof. Rongjie Lai

Normalizing Flows Optimal Transport implementation on MNIST Dataset

WGAN and Monge Map implementation on MNIST Dataset

Project in Neural Networks and Dynamical Systems

Purdue University, West Lafayette

November 2024–December 2024

MA59500MM: Introduction to Mathematical Modeling | Instructor: Prof. Alexandria Volkening

Physics-Informed Neural Networks (PINNs) for Hurricane Trajectory Prediction

Project in Biotechnology

MIT Lincoln Laboratory Summer Research Program

June 2023–July 2023

2023 MIT Lincoln Laboratory Intern Innovative Idea Challenge (I³C) (3rd Place Project Winner)

Supervised by: Ryan Burrow and Ashok Kumar, MIT Lincoln Laboratory

SKINS: Skin-growth boosting and Intra-absorptive Solution bandages

Awards

Fellowships, Scholarships and Prizes

2025 Universities Research Association-Sandia National Labs Graduate Summer Fellowship

May 2025–August 2025

2023 National GEM Consortium PhD Science Fellowship

August 2023–May 2024

- Purdue University Department of Mathematics Sponsorship

August 2023–May 2024

- MIT Lincoln Laboratory Employer Sponsorship (Internship)

May 2023–August 2023

2023 MIT Lincoln Laboratory I³C 3rd Place Research Proposal Prize

July 2023

2022 Evertec Inc. STEM Scholarship

October 2022

Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship

August 2019–December 2022

Merits and Honors

2023 Ford Foundation Predoctoral Fellowship Honorable Mention

March 2023

2022 Hispanic Scholarship Fund Scholar

June 2022

National Math Alliance Predoctoral Scholar

November 2021

UPRM Faculty of Arts and Sciences Honor Roll

August 2018–May 2023

National Trig-Star Math Competition, 16th Overall Finalist

June 2017

Eagle Scout Rank, with 2 Silver Palms

May 2017

Papers and Articles (The asterisk symbol (*) denotes alphabetical order authorship.)

Research Articles and Preprints:

[1] S. Polk, E.J. Pabon-Cancel, R. Paleja, K. Chestnut-Chang, R. Jensen and M. Ramirez.

Unsupervised Behavior Inference from Human Action Sequences (UNBIAS).

2024 IEEE Conference on Games (CoG), Milan, Italy, 2024, pp. 1-8.

DOI: <https://doi.org/10.1109/CoG60054.2024.10645587>

[2] *A. Allen, E.J. Pabon-Cancel, F. Piñero-Gonzalez and L. Polanco.

Improving the Dimension Bound of Hermitian-Lifted Codes.

arXiv: <https://arxiv.org/abs/2302.01557>

[3] *D. Chen, P.E. Harris, J. Carlos Martinez Mori, E.J. Pabon-Cancel and G. Sargent.

Permutation Invariant Parking Assortments.

Enumerative Combinatorics and Applications, **4:1**, 1-25 (2024). #S2R4.

DOI: <https://doi.org/10.54550/ECA2024V4S1R4>

[4] *I. Byrne, N. Dodson, R. Lynch, E.J. Pabon-Cancel and F. Piñero-Gonzalez.

Improving the minimum distance bound of Trace Goppa codes.

Designs, Codes and Cryptography. **91**, 2649–2663 (2023).

DOI: <https://doi.org/10.1007/s10623-023-01216-6>

Contributions to the profession:

- [1] *P.E. Harris, Z. Markman, L. Martinez, A. Mock, E.J. Pabón-Cancel, A. Verga, and S. Wang.
A Model for a One-Hour Workshop on Mentoring.
MAA Focus, **43**(1), 18-21 (2023).
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Service and Outreach

Academic Service

- Universities Research Association
URA STEM Ambassadors Program
University Ambassador (Purdue University) March 2026–present
West Lafayette, Indiana

Teaching and Grading

- MA 59500MB: Mathematical Biology (Grading) January 2026–May 2026
- MA 32500: History of Mathematics (Grading) January 2026–May 2026
- MA 26100 REC: Multivariate Calculus Recitation (Teaching) August 2025–December 2025
January 2025–May 2025
August 2024–December 2024
- MA 13900: Mathematics for Elementary Teachers III (Grading) June 2024–August 2024

Student Associations

- Purdue SACNAS Chapter
Purdue University
Member August 2025–May 2026
West Lafayette, Indiana
 - Pythagorum
University of Puerto Rico, Mayagüez Campus
Co-founder & Vice-President August 2022–December 2022
Mayagüez, Puerto Rico
 - Society of Physics Students, UPRM Chapter
University of Puerto Rico, Mayagüez Campus
Committee Assistant August 2018–December 2022
Mayagüez, Puerto Rico
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Poster Sessions, Panels, Presentations and Conferences

- Purdue University Student Computational and Applied Mathematics Seminar
Helen B. Schleman Hall, Purdue University 27 March 2026
West Lafayette, Indiana
Panel: Internships and Job Opportunities Beyond Academia
- Purdue Graduate Student Government Research-O-Rama
West Lafayette Public Library 21 February 2026
West Lafayette, Indiana
Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Prospective Student Visit Weekend Graduate Student Research Expo.
Mathematical Sciences Building, Purdue University 20 February 2026
West Lafayette, Indiana
Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Student Computational and Applied Mathematics Seminar
Helen B. Schleman Hall, Purdue University 13 February 2026
West Lafayette, Indiana
Presentation: Data-Driven Closure Models (DDCMs)
- URA-Sandia Graduate Summer Fellowship Lightning Talk
Presentation: Data-Driven Closure Models (DDCMs) 6 August 2025
Virtual Seminar
- Sandia National Laboratories CA SIP Intern Symposium
Auditorium, Sandia National Laboratories-Livermore 5 August 2025
Livermore, California
Poster: Data-Driven Closure Models (DDCMs)
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce)
Invited REU Seminar: My Story & Permutation-Invariant Parking Assortments 18 July 2025
Virtual Seminar

- Purdue University Student Commutative Algebra Seminar
Helen B. Schleman Hall, Purdue University
Presentation: Results in $\tau_{(n)}$ -factorizations and $\tau_{(n)}$ -primes. 18 November 2024
West Lafayette, Indiana
- Purdue University Student Math History Seminar
Lawson Computer Science Building, Purdue University
Presentation: Testimonios: Stories of Latinos and Hispanics in Mathematics 9 September 2024
West Lafayette, Indiana
- Underrepresented Students in Topology and Algebra Research Symposium 2024
University of Iowa 20-21 April 2024
Iowa City, Iowa
- 2023 MIT Lincoln Lab Intern Innovative Idea Challenge
MIT Lincoln Laboratory Auditorium 14, 21 July 2023
Lexington, Massachusetts
Poster: Skin-Absorptive and Skin-Growth Boosting Bandages
Presentation: SKINS: Skin-growth boosting and Intra-absorptive Solution Bandages
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce)
Invited REU Seminar Talk: Graduate School: Application tips and advice 7 July 2023
Virtual Seminar
- 2023 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting
University of Puerto Rico at Bayamón, Sponsored by PR-LSAMP 29 April 2023
Bayamón, Puerto Rico
Presentation: Properties of $\tau_{(n)}$ -primes
- 38th Interuniversity Mathematical Sciences Research Seminar
University of Puerto Rico, Mayagüez Campus 24-25 February 2023
Mayagüez, Puerto Rico
Presentation: Permutation Invariant Parking Assortments
- 2023 AAAS Emerging Researchers National Conference in STEM
Omni Shoreham Hotel 9-11 February 2023
Washington, District of Columbia
Poster: Permutation Invariant Parking Functions with cars of assorted lengths
- Joint Mathematics Meetings 2023
John B. Hynes Veterans Memorial Convention Center 4-7 January 2023
Boston, Massachusetts
Poster: Permutation Invariant Parking Functions with cars of assorted lengths
Presentation: Permutation Invariant Parking Functions with Cars of Arbitrary Lengths
- Field of Dreams Conference 2022
The Graduate Hotel, University of Minnesota-Twin Cities 4-6 November 2022
Minneapolis, Minnesota
- 2022 SACNAS National Diversity in STEM Conference
Pedro Roselló Convention Center 27-29 October 2022
San Juan, Puerto Rico
Poster: The Study of $\tau_{(n)}$ -primes
- 2022 Gulf Coast Undergraduate Research Symposium
William Marsh Rice University 8-9 October 2022
Houston, Texas
Presentation: Properties of $\tau_{(n)}$ -primes
- Summer@ICERM 2022: Computational Combinatorics
Institute for Computational and Experimental Research in Mathematics 3 August 2022
Providence, Rhode Island
Presentation: On Permutation-Invariant Parking Sequences
- 2022 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting
University of Puerto Rico at Humacao 9 April 2022
Humacao, Puerto Rico
Presentation: The Study of $\tau_{(n)}$ -primes
- Joint Mathematics Meetings 2022
Poster: Improving Bounds of Hermitian-Lifted Codes 6-9 April 2022
Virtual Conference
- 37th Interuniversity Mathematical Sciences Research Seminar
Poster: The Study of $\tau_{(n)}$ -primes 25-26 February 2022
Virtual Conference
Presentation: Improving Bounds of Hermitian-Lifted Codes
- 2021 Math REU Conference@Clemson University
Presentation: Improved Hermitian-Lifted Codes 19 July 2021
Virtual Conference

- 2021 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting
Presentation: The Study of $\tau_{(n)}$ -atoms
23-24 April 2021
Virtual Conference
- 35th Interuniversity Mathematical Sciences Research Seminar
University of Puerto Rico at Cayey
Poster: The Study of $\tau_{(n)}$ -atoms
6-7 March 2020
Cayey, Puerto Rico

Academics and Coursework

Purdue University

Coursework (Graduate):

MA 61500: Numerical Methods for Partial Differential Equations	January 2026–May 2026
MA 53200: Elements of Stochastic Processes	January 2026–May 2026
MA 59800ZAT: Hamiltonian Dynamics	January 2026–May 2026
MA 55400: Linear Algebra	August 2025–December 2025
MA 51900: Introduction to Probability	August 2025–December 2025
MA 57300: Numerical Solutions of Ordinary Differential Equations	August 2025–December 2025
MA 59500OT: Computational Optimal Transport and Deep Generative Models	January 2025–May 2025
MA 59800ZY: Topics in Dynamical Systems (Bifurcation Theory)	January 2025–May 2025
MA 54600: Introduction to Functional Analysis	January 2025–May 2025
MA 59500AFF: Analytic Theory of Function Fields	August 2024–December 2024
MA 59500MM: Introduction to Mathematical Modeling	August 2024–December 2024
MA 54300: Introduction to Ordinary Differential Equations and Dynamical Systems	January 2024–May 2024
MA 54400: Real Analysis and Measure Theory	January 2024–May 2024
MA 55300: Introduction to Abstract Algebra	August 2023–December 2023
MA 53000: Functions of a Complex Variable I	August 2023–December 2023

University of Puerto Rico, Mayagüez Campus

Coursework (Graduate):

MATE 6101: Number Theory I	August 2022–December 2022
MATE 5150: Linear Algebra	January 2021–May 2021

Coursework (Undergraduate):

MATE 4000: Elements of Topology	August 2022–December 2022
MATE 4052: Advanced Calculus II	January 2022–May 2022
MATE 4051: Advanced Calculus I	August 2021–December 2021
MATE 4061: Numerical Analysis I	August 2021–December 2021
MATE 4021: Introduction to Mathematical Logic	August 2021–December 2021
MATE 4010: Introduction to Complex Variables	January 2021–May 2021
ESMA 4001: Mathematical Statistics I	January 2021–May 2021
MATE 4020: Partial Differential Equations and Fourier Series	August 2020–December 2020
MATE 4008: Introduction to Algebraic Structures	August 2020–December 2020
MATE 3040: Theory of Numbers	August 2020–December 2020
MATE 4009: Ordinary Differential Equations	June 2020–July 2020
MATE 4031: Introduction to Linear Algebra	January 2020–May 2020
COMP 3010: Introduction to Computer Programming I	January 2020–May 2020
MATE 3063: Calculus III	January 2020–May 2020
MATE 3032: Calculus II	August 2019–December 2019
MATE 3020: Introduction to Mathematical Foundations	August 2019–December 2019
MATE 3031: Calculus I	June 2019–July 2019
MATE 3172: Precalculus II	January 2019–May 2019
MATE 3171: Precalculus I	August 2018–December 2018